Meehir Prabhakar

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Education

VIT-AP University — India
Bachelor of Technology - BTech, Computer Science and Engineering — *Aug. 2023 – Expected Jun 2027*

Bhatnagar International School — India
PCM (Physics, Chemistry, Mathematics)

Convent of Rani Jhansi — India
Primary Education

Professional Experience & Applied Research

TechtoGreen Drone & Robotics Private Limited — Amaravati, Andhra Pradesh, India
AI/ML and Computer Vision Intern — *Dec. 2024 – May 2026*

- **PROJECT: Autonomous Precision Agriculture & Telemetry**
- Engineered a real-time crop disease detection pipeline using YOLO and OpenCV, processing live drone video feeds to identify structural anomalies and infections across 7 commercial crops.
- Integrated MAVLink telemetry to synchronize model inference with drone GPS coordinates, mapping high-confidence infection clusters onto interactive Folium and Streamlit dashboards.
- Developed an offline Text-to-Speech engine using VITS (Sherpa-ONNX) to narrate region-specific pesticide treatments and agronomic recommendations for field operators.
- Designed a multi-threaded PyQt6 graphical interface enabling real-time confidence tuning, sensor monitoring, and seamless model switching during flight operations.
- Developed a synchronized video and telemetry recording thread, coupled with an automated batch-inference engine to run YOLO disease detection on pre-recorded drone flights.
- Engineered a smart serial port detection algorithm to automatically identify and connect ESP32/Arduino telemetry modules, ensuring seamless field deployments.
- Implemented a dynamic mock telemetry injector for hardware-free UI testing and adapted inference pipelines to optimize confidence thresholds for high-altitude crop analysis.

- **PROJECT: Automated Agronomic Reporting Engine**
- Engineered a dynamic PDF report generation system within a FastAPI backend using the fpdf2 library to summarize YOLO-detected crop diseases, processing metrics, and localized treatments.
- Implemented a robust internationalization (i18n) pipeline to render agricultural reports in both English and Telugu, utilizing custom dictionary mappings for precise translation of crop names and pathology terms.
- Integrated text shaping logic via uharfbuzz and custom Noto Sans typography fallback mechanisms to guarantee the accurate rendering of complex regional scripts and glyphs in cross-platform PDF exports.

- **PROJECT: Warehouse Inventory & Occupational Safety**
- Developed a real-time box and SKU counting system utilizing custom-trained YOLO models (nSKU/BoxE) to automate inventory management from live camera feeds.
- Engineered a bespoke Non-Maximum Suppression (NMS) algorithm using Intersection over Union (IoU) metrics to eliminate overlapping bounding boxes and ensure high-precision counting.
- Designed a frame-history tracking mechanism to filter transient false positives in live video, automatically logging verified count changes for stable inventory tracking.
- Developed a real-time worker safety pipeline combining YOLOv8 person detection and MediaPipe Pose to evaluate ergonomic posture by computing spinal angles from skeletal landmarks.
- Integrated a custom-trained YOLO model into the vision pipeline to verify hardhat compliance by analyzing isolated worker regions of interest.
- Engineered a dynamic collision-avoidance system deploying dual YOLO models to track personnel and forklifts, utilizing a bounding box intersection algorithm to trigger visual proximity warnings.
- Designed a video ingestion and logging architecture using OpenCV to overlay real-time posture diagnostics and helmet status, exporting timestamped footage for safety compliance auditing.

- **PROJECT: Autonomous Aquatic Surface Robot (TGWV1)**
- Engineered the primary computer vision pipeline for a water surface cleaning robot (TGWV1), deploying custom YOLO models to detect, classify, and track floating waste and wildlife obstacles.
- Interfaced a YDLIDAR G2 sensor with a custom C++ application using the YDLidar-SDK to process 360-degree point cloud data in real-time.

- Developed a spatial filtering algorithm to isolate a 30-degree forward-facing Field of View (FOV), identifying obstacles within a 1.2-meter safe zone to trigger GPIO haptic feedback.
- Built a parallel computer vision pipeline in Python using OpenCV and a custom-trained YOLO model to detect environmental hazards like fires and potholes from live webcam feeds.
- Implemented a focal-length-based spatial estimation algorithm to calculate the proximity and relative intercept angle of aquatic debris directly from 2D bounding box data.
- Developed the directional signaling logic utilizing the lgpio library to translate visual target coordinates into hardware-level motor control triggers, guiding the vessel's navigation microcontrollers.

- **PROJECT: 3D Scene Reconstruction & Vegetation Analysis**
- Developed a dual-model vision pipeline combining VGGT monocular depth estimation and YOLO instance segmentation to isolate nearby vegetation. Engineered a bitwise mask intersection algorithm to filter foliage within a 1.0-meter proximity threshold.
- Implemented a 3D point-cloud generation tool using Trimesh to convert VGGT depth predictions and camera extrinsics into interactive GLB environments. Integrated an ONNX-based sky segmentation model to filter out background noise from the reconstructed scenes.
- Created an automated OpenCV pipeline to overlay real-time camera coordinates, timestamps, and segmentation masks onto video feeds for precise geospatial tracking of overgrown bushes.

- **PROJECT: SentinelHelm Intelligent Surveillance**
- Engineered a real-time AI security platform integrating YOLO-based weapon detection and DeepFace facial recognition to identify threats in live video feeds.
- Developed a Person of Interest (POI) tracking pipeline using ArcFace embeddings and FAISS for high-speed vector similarity search, linked to an SQLite metadata database. Offloaded embedding extraction to a QThreadPool to prevent UI freezing.
- Designed a custom heuristic algorithm utilizing YOLO pose keypoints to identify fighting stances and calculate limb velocity for physical hit detection. Integrated a 5-class weapon and fist detection model to minimize false positives.
- Built a responsive Material Design interface using PyQt6 and QML (QtQuick) for live stream viewing and database management. Implemented dynamic GPU memory allocation via TensorFlow to resolve Windows-specific memory crashes.

- **PROJECT: Cross-Modal Spectral & Visual Analysis**
- Engineered a cross-modal pseudo-labeling pipeline, utilizing DBSCAN clustering on FT-IR spectra to generate chemical ground-truth labels for training visual anomaly detection models.
- Developed an unsupervised anomaly detection pipeline for FT-IR spectra, applying Savitzky-Golay filtering, SNV scatter correction, and PCA dimensionality reduction prior to Isolation Forest classification.
- Built a custom computer vision feature extractor to quantify turmeric purity, fusing 1D arrays of Hue/Saturation histograms and Local Binary Pattern (LBP) texture maps to bypass deep learning overhead.
- Trained Random Forest and One-Class SVM classifiers on the extracted visual features, implementing SMOTE for minority class synthesis and PCA for component compression.

- **PROJECT: Visual SLAM & Odometry Engine**
- Engineered a synchronized data capture pipeline integrating stereo webcams, an MPU9250 IMU, and NMEA GPS receivers via PySerial to generate multi-modal datasets for odometry evaluation.
- Developed a monocular VO system utilizing ORB feature extraction and FLANN-based matching, implementing Essential matrix decomposition and point triangulation to mathematically recover relative camera scale.
- Built a complete visual SLAM architecture featuring StereoSGBM disparity mapping, FAST corner detection, and Lucas-Kanade (KLT) optical flow tracking to process binocular video streams.
- Implemented Levenberg-Marquardt optimization to minimize 3D-2D reprojection residuals for accurate 6-DOF camera pose estimation, reconstructing and exporting global 3D point cloud environments.

The Machine Learning CLUB VITAP Amaravati, Andhra Pradesh, India

Technical & Project Team Member *Oct. 2023 – May. 2025*

- **Game Engine Architecture:** Engineered a 3D survival game (PAC 3D) in Godot Engine using GDScript, implementing custom physical environments, ambient audio systems, and state-persistent UI overlays.
- **AI & Pathfinding:** Programmed dynamic enemy AI utilizing Godot's NavigationMesh for spatial pathfinding, featuring smooth animation blending and progressive difficulty scaling algorithms.
- **Gameplay Systems:** Designed real-time player HUD components, including a dynamic minimap, pause state management, and modular scene transitions. (GitHub Repository)
- Contributed to club operations and open-source project development initiatives utilizing machine learning and programming tools.

Projects

Project Chakshu (AI SIEM & MDR Pipeline) | *Python, Scikit-Learn, SHAP, LIME* Jan. 2026 – May 2026

- Engineered an AI-driven SIEM/MDR framework that normalizes heterogeneous telemetry across network flows, web apps, and OS logs (Loghub, LANL, CICIDS) into a standardized JSON schema.
- Deployed layer-specific ML models, utilizing Isolation Forests for L3/L4 network anomaly detection and One-Class SVMs with TF-IDF character N-grams for L7 web and auth threat isolation.
- Built a central Fusion Engine utilizing a 60-second temporal sliding window for cross-layer attack correlation, integrating SHAP and LIME for explainable threat attribution.
- Hosted on GitHub: github.com/Meehir5907/Chakshu

Autonomous Sewage Maintenance Submarine | *Python, YOLOv8n, NCNN, Open3D, RPi 5* Jul. 2024 – Nov. 2025

- Engineered a neutrally buoyant autonomous submarine for deep sewage inspection to replace manual scavenging, utilizing an isolated 11.1V dual-battery architecture to prevent CPU brownouts.
- Deployed an NCNN-optimized YOLOv8n instance segmentation model on a headless Raspberry Pi 5 to identify and map structural cracks in real-time at 3.0 FPS.
- Designed a mechanical SONAR imaging system (DYP-A08) utilizing Open3D noise reduction and Matplotlib bicubic interpolation to generate spatial depth heatmaps for zero-visibility navigation.
- Programmed 3D propulsion and robotic arm actuation via `lgpio`, controlling RS2205 brushless motors via 30A ESCs and a PCA9685 PWM controller.

Awards & Achievements

1st Place (Dept. of AI/DS), Innovative Model: Secured 1st place at Siddhartha Academy of Higher Education Innovation Day (Oct. 2025) in department of Artificial Intelligence / Data Science.
AI Hackathon 2025: AIMERS Letter of Appreciation (Feb. 2025).
Ideathon 2K25: Innovation Showcase Participant at Chalapathi Institute of Engineering & Technology (Mar. 2025).

Technical Skills

Languages: Python, C/C++, SQL, Go, Bash/Zsh, Lua, Kotlin, GDScript
AI, ML & Vision: PyTorch, TensorFlow, Keras, OpenCV, YOLO, MediaPipe, FAISS, Transformers, SHAP, LIME
Systems & Tools: Linux (KVM, Kernel Tuning, Cron), Git/GitHub, FastAPI, QML, Volatility 3, AVML
Hardware & Edge: ESP32, Arduino, LiDAR (YDLIDAR), MAVLink Telemetry, Raspberry Pi
Core Domains: Computer Vision, Visual SLAM, SIEM/MDR Pipelines, Memory Forensics, Unsupervised Learning